



**The Faculty of Information and Communication Technology
Mahidol University**

661_ITCS241

Database Management Systems

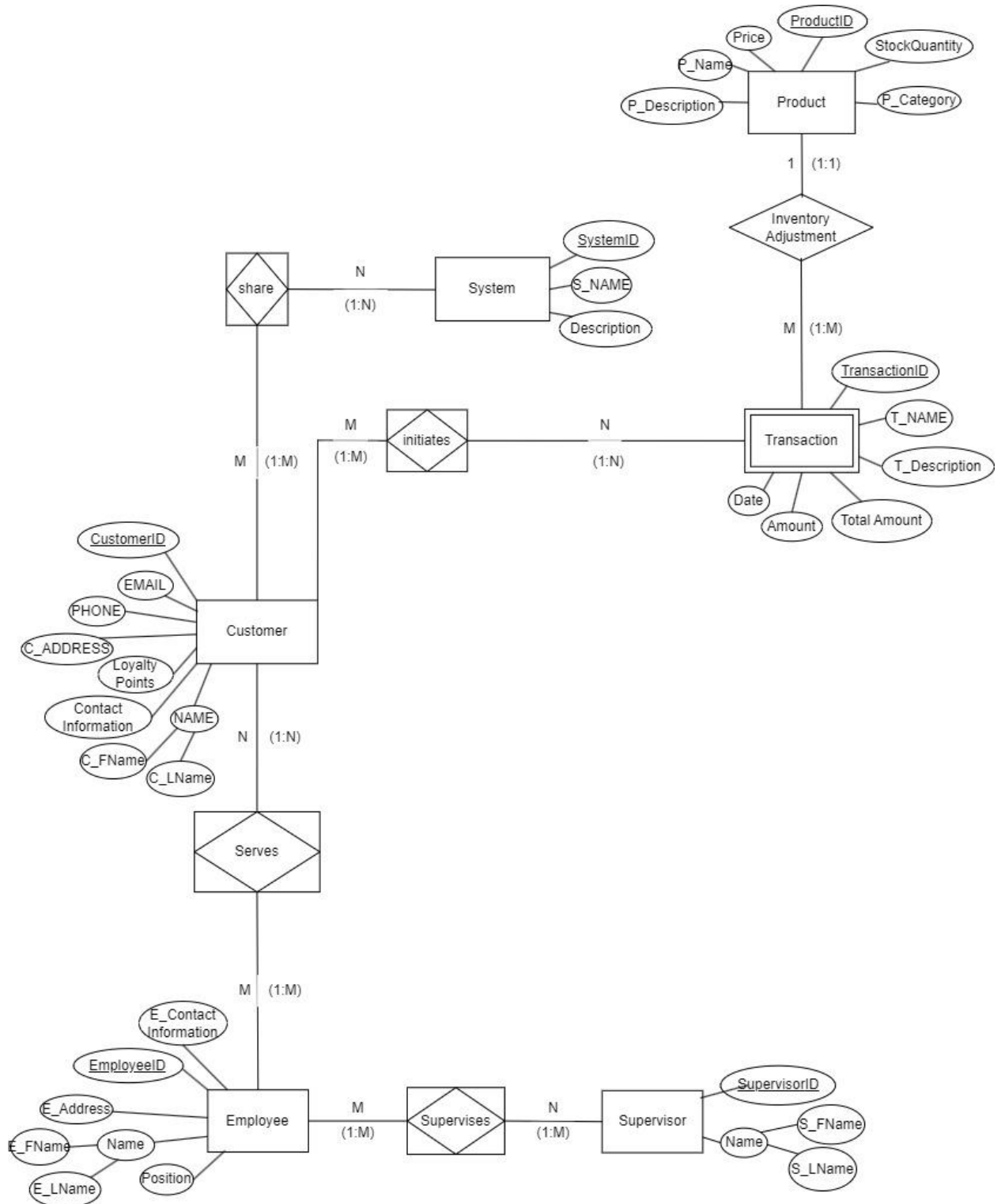
**Project Phase II Final Submission
Semester 1, 2023**

MEMBER

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Revised ERD/EERD



8 Steps transform ERD to Relation Model

Steps 1 - Transforming regular entity types

Customer	<u>CustomerID</u>	C_FNAME	C_LNAME	EMAIL	PHONE	C_ADDRESS	LOYAL POINTS	C_CONTACT INFORMATION
Employee	<u>EmployeeID</u>	E_FNAME	E_LNAME	POSITION	E_ADDRESS	E_CONTACT INFORMATION		
Supervisor	<u>SupervisorID</u>	S_FNAME	S_LNAME					
System	<u>SystemID</u>	S_NAME	S_Description					
Product	<u>ProductID</u>	P_NAME	P_Description	Price	StockQuantity	P_Category		

Steps 2 - Transforming weak entities

Transaction	<u>TransactionID</u>	T_NAME	T_Description	Total Amount	Amount	Date
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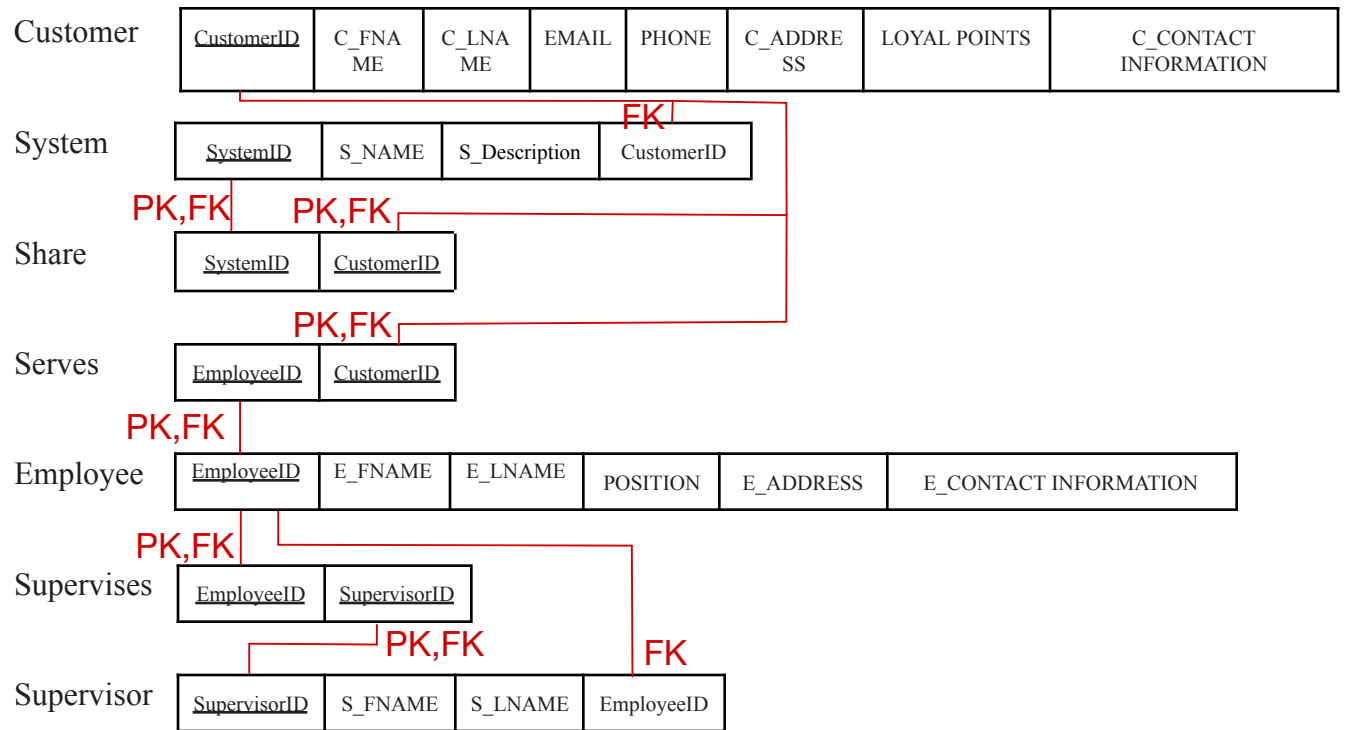
Steps 3 - No Transforming binary 1:1 relationship types

Steps 4 - Transforming binary 1:N relationship types

Product	<u>ProductID</u>	NAME	Description	Price	StockQuantity	Category		
Transaction	<u>TransactionID</u>	T_NAME	T_Description	Total Amount	Amount	Date	ProductID	CustomerID

FK

Steps 5 - Transforming binary M:N relationship types

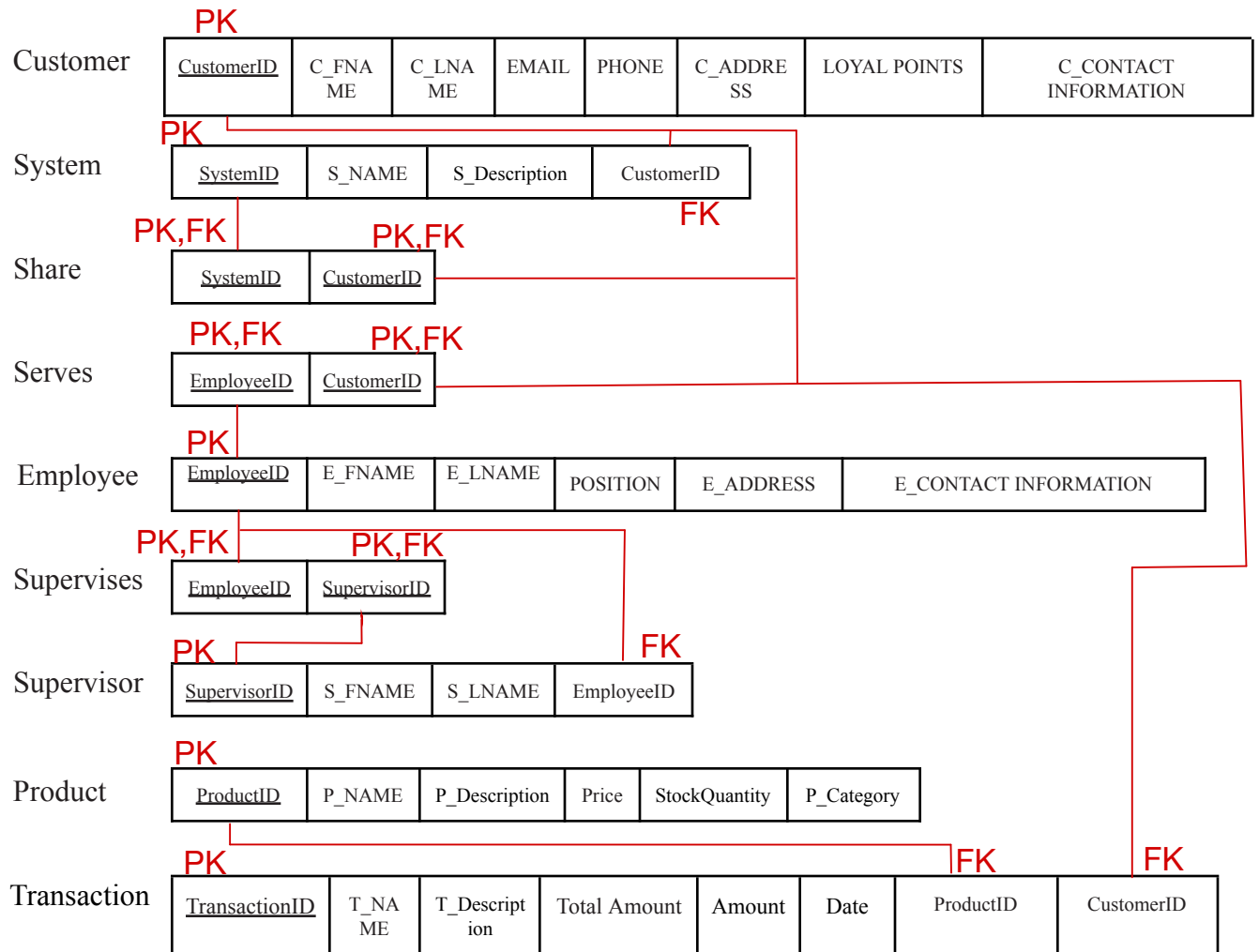


Steps 6 - No Transforming multi-valued attribute A in relation K

Steps 7 - No N-ary relationship type

Steps 8 - No Transforming specialization or generalization

Final Schema



Attr. Name	Contents	Types	Nullable	Key	FK Ref.Table	Example Data
Customer						
CustomerID	Customer's ID	INT		PK		10
C_FNAME	Customer's first name	VARCHAR (50)	Y			Tanasub
C_LNAME	Customer's last name	VARCHAR (50)	Y			Rattanaharirak
EMAIL	Email address of customer	VARCHAR (100)	Y			tanasub.rat@gmail.com
PHONE	Phone number of customer	VARCHAR (10)				0878987111
C_ADDRESS	Customer's address	VARCHAR (255)	Y			123 Sukhumvit Rd, BKK
LOYAL POINTS	Points to receive rewards for customer	INT				312
C_CONTACTINFORMATION	Customer's Facebook for contact	VARCHAR (100)				Tanasub_facebook

System						
SystemID	System's ID	INT		PK		91230
S_NAME	System name	VARCHAR(50)	Y			Loyal System
Description	Description of each system	VARCHAR(255)				Reward for customer
CustomerID	Customer's ID	INT		FK	CustomerID (Customer)	10

Share						
SystemID	System's ID	INT		PK,FK	SystemID (System)	91230
CustomerID	Customer's ID	INT		PK,FK	CustomerID (Customer)	10

Serves						
EmployeeID	Employee's ID	INT		PK,FK	EmployeeID (Employee)	112
CustomerID	Customer's ID	INT		PK,FK	CustomerID (Customer)	91230

Employee						
EmployeeID	Employee's ID	INT		PK		112
E_FNAME	Employee's first name	VARCHAR(50)	Y			Phakphop
E_LNAME	Employee's last name	VARCHAR(50)	Y			Phunphoem
POSITION	Role of employee	VARCHAR(50)				Cashier
E_ADDRESS	Employee's email	VARCHAR(100)				Phakphop.ph@gmail.com
E_CONTACT INFORMATION	Employee's Facebook for contact	VARCHAR(100)				Phakphop_Phu_Facebook

Supervises						
EmployeeID	Employee's ID	INT		PK,FK	EmployeeID (Employee)	112
SupervisorID	Supervisor's ID	INT		PK,FK	SupervisorID (Supervisor)	312300

Supervisor						
SupervisorID	Supervisor's ID	INT		PK		312300
S_FNAME	Supervisor's first name	VARCHAR(50)	Y			Poottapol
S_LNAME	Supervisor's last name	VARCHAR(50)	Y			Poonna
EmployeeID	Head of this employee's ID	INT		FK	EmployeeID (Employee)	112

Product						
ProductID	Product's ID	INT		PK		1121
P_NAME	Name of the product	VARCHAR(50)				Supasri_snack
P_Description	Description of the product	VARCHAR(255)	Y			Premium chocolate snack bar
Price	Price of each product	DECIMAL(10,2)				50.00
StockQuantity	Quantity of each product	INT				120
P_Category	Type of product	VARCHAR(50)	Y			Snack

Transaction						
TransactionID	Transaction's ID	INT		PK		11312
T_NAME	Transaction's name	VARCHAR(50)				Purchase1
T_Description	Description of the product	VARCHAR(255)	Y			Grocery shopping
Total Amount	Price of all product in cart	DECIMAL(10,2)				300
Amount	Price of each product	DECIMAL(10,2)				300
Date	Date of transaction	DATE				2023-11-18
ProductID	Product's ID that customer purchase	INT		FK	ProductID (Product)	1121
CustomerID	ID of the customer who make this transaction	INT		FK	CustomerID (Customer)	10

Data Dictionary

Database Query

6588083 Tanasub Rattanaharirak

Basic Queries

1. Report: Customer firstname, lastname and email.

Objective: To retrieve all customer name and email.

SQL Command:

```
SELECT CONCAT(C_FNAME," ",C_LNAME) as "Customer Name", EMAIL  
FROM Customer;
```

Query Result:

	Customer Name	EMAIL
▶	Tanasub Rattanaharirak	tanasub.rat@gmail.com
	Poottapol Paannna	poottapol.ponna@gmail.com
	Phakphup Phoonphoem	phakphup.phoonphoem@gmail.com
	Kasama Kiatapakul	kasamatransaction.kiat@gmail.com
	Supasri Seekane	supasr.seekane@gmail.com
	Nantipat Lnwza	nantipat.Lnw@gmail.com
	Pimmy Nekognum	pimmy,nego@gmail.com
	Pathu Sudlhor	pathu.sudlhor@gmail.com
	Napat Zaza	napat.zaza@gmail.com
	Ratchaneekorn Suaymak	ratchaneekorn.suay@gmail.com
	Thawanthong Lhormak	thawanthong.lhor@gmail.com

2. Report: Transaction name and amount.

Objective: To retrieve Transactions name and amount greater or equal to 400.

SQL Command:

```
SELECT T_NAME, Amount  
FROM Transaction  
Where amount >= 400;
```

Query Result:

	T_NAME	Amount
▶	Order7	400.00
	Purchase8	450.00
	Shopping9	500.00
	Order10	550.00
	Shopping11	600.00

3. Report: ProductID,product name and description of product.
 Objective: To retrieve productID,product name and description.
 SQL Command:
 SELECT ProductID, P_NAME, P_Description
 FROM Product;

Query Result:

	ProductID	P_NAME	P_Description
▶	1121	Supasri snack	Premium chocolate snack bar
	1122	Taro Cupcake	Delicious and sweet Taro-flavored cupcake
	1123	Bubble Milk Tea	Sweet and soft bubble milk tea
	1124	Traditional Thai Clothing	Handwoven traditional Thai clothing made from ...
	1125	Natural Ink Cartridge	Ink cartridge made from Thai herbs
	1126	Genuine Leather Bag	Genuine leather bag made from selected animal ...
	1127	Shockproof Hat	Vibrant shockproof hat for sports
	1128	Thai Language Book	Interesting Thai language book
	1129	Thai Tea Cup	Thai tea cup adorned with gold patterns
	1130	Sensory Balls	Colorful sensory balls for play
	1131	Herbal Soap	Herbal soap made from Thai herbs
	1132	Iphone 15 pro max 256...	Apple product
	1133	Ipad Pro 2022 256GB	Apple product
	1134	Macbook Air M1 512	Apple product
*	NULL	NULL	NULL

Advance Queries

4. Report: Customer firstname, last name ,transaction name and total amount .
 Objective: To find customers and their transactions.
 SQL Command:
 SELECT C.C_FNAME, C.C_LNAME, T.T_NAME, T.TotalAmount
 FROM Customer C
 JOIN Transaction T ON C.CustomerID = T.CustomerID;

Query Result:

	C_FNAME	C_LNAME	T_NAME	TotalAmount
▶	Tanasub	Rattanaharirak	Purchase1	300.00
	Poottapol	Paannna	Shopping11	1800.00
	Phakphup	Phoonphoem	Order7	1200.00
	Kasama	Kiatapakul	Purchase2	450.00
	Kasama	Kiatapakul	Purchase8	1350.00
	Supasri	Seekane	Shopping9	1500.00
	Pimmy	Nekognum	Purchase5	900.00
	Pathu	Sudlhor	Order10	1650.00
	Napat	Zaza	Shopping3	600.00
	Napat	Zaza	Shopping6	1050.00
	Thawant...	Lhormak	Order4	750.00

5. Report: CustomerID, Customer firstname and last name.

Objective: To find customers who made transactions on 2023-11-18.

SQL Command:

```
SELECT C.CustomerID, C.C_FNAME, C.C_LNAME
```

```
FROM Customer C
```

```
WHERE C.CustomerID IN (SELECT DISTINCT T.CustomerID FROM Transaction T WHERE  
T.Date = '2023-11-18');
```

Query Result:

	CustomerID	C_FNAME	C_LNAME
▶	10	Tanasub	Rattanaharirak
✱	NULL	NULL	NULL

6. Report: CustomerID, Customer firstname, last name and total transaction.

Objective: To get the total number of transactions for each customer.

SQL Command:

```
SELECT C.CustomerID, C.C_FNAME, C.C_LNAME, COUNT(T.TransactionID) AS
```

```
TotalTransactions
```

```
FROM Customer C
```

```
LEFT JOIN Transaction T ON C.CustomerID = T.CustomerID
```

```
GROUP BY C.CustomerID;
```

Query Result:

	CustomerID	C_FNAME	C_LNAME	TotalTransactions
▶	10	Tanasub	Rattanaharirak	1
	11	Poottapol	Paannna	1
	12	Phakphup	Phoonphoem	1
	13	Kasama	Kiatapakul	2
	14	Supasri	Seekane	1
	15	Nantipat	Lnwza	0
	16	Pimmy	Nekognum	1
	17	Pathu	Sudlhor	1
	18	Napat	Zaza	2
	19	Ratchaneekorn	Suaymak	0
	20	Thawanthong	Lhormak	1

7. Report: Supervisor firstname, lastname and Employee firstname, last name.

Objective: To find supervisors and their subordinates.

SQL Command:

```
SELECT S.S_FNAME, S.S_LNAME, E.E_FNAME, E.E_LNAME
FROM Supervisor S
JOIN Supervises SS ON S.SupervisorID = SS.SupervisorID
JOIN Employee E ON SS.EmployeeID = E.EmployeeID;
```

Query Result:

	S_FNAME	S_LNAME	E_FNAME	E_LNAME
►	Poottapol	Poonna	Phakphop	Phunphoem
	Tinder	Louis	Yaya	Suaysood
	John	Terry	Supanat	Muenta

8. Report: Customer firstname and last name.

Objective: To retrieve customers who purchased a ProductID 1121

SQL Command:

```
SELECT C.C_FNAME, C.C_LNAME
FROM Customer C
WHERE C.CustomerID IN (SELECT DISTINCT T.CustomerID FROM Transaction T WHERE
T.ProductID = 1121);
```

Query Result:

	C_FNAME	C_LNAME
►	Tanasub	Rattanaharirak

9. Report: Customer firstname and last name.

Objective: To find customers who purchase a snack.

SQL Command:

```
SELECT C_FNAME, C_LNAME
FROM Customer
WHERE CustomerID IN (
    SELECT DISTINCT T.CustomerID
    FROM Transaction T
    JOIN Product P ON T.ProductID = P.ProductID
    WHERE P.P_Category = 'Snack'
);
```

Query Result:

	C_FNAME	C_LNAME
►	Tanasub	Rattanaharirak

10. Report: ProductID,Product name,Product description, Price, Stock quantity, category.
Objective: To list products that have a stock quantity greater than the average stock quantity.

SQL Command:

SELECT *

WHERE StockQuantity > (

FROM Product

ORDER BY StockQuantity DESC;

Query Result:

6588079 Poottapol Poonna

11. Report: All Customer data.

SQL Command:

from customer;

Query Result:

[illegible]

Advance Queries

14.Report: CustomerID, Customer Firstname, Customer Lastname, Email, Phone, Customer Address, Loyal Point, Customer Contact Information.

Objective: Find the top 5 customers with the highest loyalty points.

SQL Command:

```
select *
```

```
from customer order by loyal_points desc limit 5;
```

Query Result:

	CustomerID	C_FNAME	C_LNAME	EMAIL	PHONE	C_ADDRESS	LOYAL_POINTS	C_CONTACTINFORMATION
▶	10	Tanasub	Rattanaharirak	tanassub.rat@gmail.com	0878987111	123 Sukhumvit Rd, BKK	312	Tanasub_facebook
	18	Napat	Zaza	napat.zaza@gmail.com	0923456789	321/8 Thong Lor Street, Phang Nga	250	Napat_facebook
	14	Supasri	Seekane	supasr.seekane@gmail.com	0954321987	123/45 Charoen Krung Street, Pattaya	220	Supasri_facebook
	17	Pathu	Sudhor	pathu.sudhor@gmail.com	0943219876	777/33 Chang Klan Road, Krabi	200	Pathu_facebook
	13	Kasama	Kiatapakul	kasama.kiat@gmail.com	0898765432	78 Tha Phae Road, Phuket	180	Kasama_facebook
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

15.Report: Product Category, Total Revenue

Objective: Calculate the total revenue from each product category.

SQL Command:

```
SELECT P.P_Category, SUM(T.Amount) AS TotalRevenue
```

```
FROM Product P
```

```
JOIN Transaction T ON P.ProductID = T.ProductID
```

```
GROUP BY P.P_Category;
```

Query Result:

	P_Category	TotalRevenue
▶	Snack	300.00
	Thai Desserts	150.00
	Beverages	200.00
	Fashion	600.00
	Office Supplies	300.00
	Sports	400.00
	Books	450.00
	Home Essentials	1100.00
	Toys	550.00

16.Report: ProductID,Product Name,Product Description,Price,Stock Quantity,Product Category

Objective: What product that customer “Poottapol” purchase.

SQL Command:

```
SELECT P.*  
FROM Product P  
JOIN Transaction T ON P.ProductID = T.ProductID  
JOIN Customer C ON T.CustomerID = C.CustomerID  
WHERE C.C_FNAME = 'Poottapol';
```

Query Result:

	ProductID	P_NAME	P_Description	Price	StockQuantity	P_Category
►	1131	Herbal Soap	Herbal soap made from Thai herbs	30.00	90	Home Essentials

17.Report: Product Category,Average Price

Objective: Find the average price of products in each category and order in ascending order.

SQL Command:

```
SELECT p.p_category, AVG(p.price) AS avgprice  
FROM product p  
GROUP BY p.p_category  
order by AVG(p.price) asc;
```

Query Result:

	p_category	avgprice
►	Toys	15.000000
	Books	25.000000
	Thai Desserts	35.000000
	Home Essentials	35.000000
	Beverages	45.000000
	Sports	60.000000
	Office Supplies	80.000000
	Snack	85.000000
	Fashion	135.000000

18.Report: CustomerID,Customer Firstname,Customer Lastname

Objective: Find customer who have not made any transactions

SQL Command:

```
SELECT C.CustomerID, C.C_FNAME, C.C_LNAME  
FROM Customer C  
LEFT JOIN Transaction T ON C.CustomerID = T.CustomerID  
WHERE T.TransactionID IS NULL;
```

Query Result:

	CustomerID	C_FNAME	C_LNAME
►	15	Nantipat	Lnwza
	19	Ratchaneekorn	Suaymak

19.Report: CustomerID, Customer Firstname, Customer Lastname, Transaction Count

Objective: List the customer with the highest transaction in descending order.

SQL Command:

```
SELECT c.customerid, c.c_FNAME, c.c_LNAME, COUNT(T.TransactionID) AS
TransactionCount
FROM customer c
LEFT JOIN Transaction T ON c.customerID = t.customerID
GROUP BY c.customerID, c.c_FNAME, c.c_LNAME
ORDER BY TransactionCount DESC;
```

Query Result:

	customerid	c_FNAME	c_LNAME	TransactionCount
►	13	Kasama	Kiatapakul	2
	18	Napat	Zaza	2
	10	Tanasub	Rattanaharirak	1
	11	Poottapol	Paannna	1
	12	Phakphup	Phoonphoem	1
	14	Supasri	Seekane	1
	16	Pimmy	Nekognum	1
	17	Pathu	Sudlhor	1
	20	Thawanthong	Lhormak	1
	15	Nantipat	Lnwza	0
	19	Ratchaneekorn	Suaymak	0

20.Report: CustomerID, Customer Firstname, Customer Lastname, Transaction Count, Average Transaction Count

Objective: Find the average number of transaction per customer.

SQL Command:

```
SELECT c.customerID, c.c_FNAME, c.c_LNAME, COUNT(T.TransactionID) AS
TransactionCount, AVG(T.Amount) AS AvgTransactionAmount
FROM customer c
LEFT JOIN Transaction T ON c.customerID = T.customerID
GROUP BY c.customerID, c.c_FNAME, c.c_LNAME;
```

Query Result:

	customerID	c_FNAME	c_LNAME	TransactionCount	AvgTransactionAmount
►	10	Tanasub	Rattanaharirak	1	300.000000
	11	Poottapol	Paannna	1	600.000000
	12	Phakphup	Phoonphoem	1	400.000000
	13	Kasama	Kiatapakul	2	300.000000
	14	Supasri	Seekane	1	500.000000
	15	Nantipat	Lnwza	0	NULL
	16	Pimmy	Nekognum	1	300.000000
	17	Pathu	Sudlhor	1	550.000000
	18	Napat	Zaza	2	275.000000
	19	Ratchaneekorn	Suaymak	0	NULL
	20	Thawanthong	Lhormak	1	250.000000

6588072 Phakphop Phunphoem

Basic Queries

21.Report: Supervisors Firstname,Supervisors Lastname

Objective: Show all supervisors

SQL Command:

```
SELECT S_FNAME, S_LNAME  
FROM Supervisor;
```

Query Report:

	S_FNAME	S_LNAME
►	Poottapol	Poonna
	Tinder	Louis
	John	Terry

22.Report: ProductID,Product Name,Product Description,Price,Stock Quantity,Product Category

Objective: Show all product with stock less than 20.

SQL Command:

```
SELECT *  
FROM Product  
WHERE StockQuantity < 20;
```

Query Report:

	ProductID	P_NAME	P_Description	Price	StockQuantity	P_Category
►	1132	Iphone 15 pro max 256GB	Apple product	999.99	10	Snack
	1133	Ipad Pro 2022 256GB	Apple product	1299.99	10	Snack
	1134	Macbook Air M1 512	Apple product	1599.99	10	Snack

23.Report: TransactionID,Transaction Name,Transaction Description,Total Amount,Amount,Date,ProductID,CustomerID

Objective: List a transaction that is online product order only.

SQL Command:

```
SELECT *  
FROM Transaction  
WHERE T_Description = 'Online product order';
```

Query Result:

	TransactionID	T_NAME	T_Description	TotalAmount	Amount	Date	ProductID	CustomerID
►	11315	Order4	Online product order	750.00	250.00	2023-11-21	1124	20
	11318	Order7	Online product order	1200.00	400.00	2023-11-24	1127	12
	11321	Order10	Online product order	1650.00	550.00	2023-11-27	1130	17

Advance Queries

24.Report: CustomerID, Customer Name, Total Revenue

Objective: Calculate the total revenue for each customer.

SQL Command:

```
SELECT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS CustomerName,  
SUM(T.Amount) AS TotalRevenue  
FROM Customer C  
JOIN Transaction T ON C.CustomerID = T.CustomerID  
GROUP BY C.CustomerID, CustomerName;
```

Query Result:

	CustomerID	CustomerName	TotalRevenue
►	10	Tanasub Rattanaharirak	300.00
	11	Pootapol Paannna	600.00
	12	Phakphup Phoonphoem	400.00
	13	Kasama Kiatapakul	600.00
	14	Supasri Seekane	500.00
	16	Pimmy Nekognum	300.00
	17	Pathu Sudlhor	550.00
	18	Napat Zaza	550.00
	20	Thawanthong Lhormak	250.00

25.Report: Product Category, Transaction Count

Objective: Find the highest transaction count.

SQL Command:

```
SELECT P.P_Category, COUNT(T.TransactionID) AS TransactionCount  
FROM Product P  
LEFT JOIN Transaction T ON P.ProductID = T.ProductID  
GROUP BY P.P_Category  
ORDER BY TransactionCount DESC  
LIMIT 1;
```

Query Result:

	P_Category	TransactionCount
►	Fashion	2

26.Report: CustomerID,Customer Name.

Objective: Identify customers who made transactions with multiple payment.

SQL Command:

```
SELECT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS CustomerName
FROM Customer C
JOIN (
    SELECT CustomerID
    FROM Transaction
    GROUP BY CustomerID
    HAVING COUNT(DISTINCT T_Description) > 1
) MultiPayment ON C.CustomerID = MultiPayment.CustomerID;
```

Query Result:

	CustomerID	CustomerName
▶	13	Kasama Kiatapakul
	18	Napat Zaza

27.Report: CustomerID,Customer Name,Latest Transaction Date

Objective: Show the latest transaction for each customer.

SQL Command:

```
SELECT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS CustomerName,
MAX(T.Date) AS LatestTransactionDate
FROM Customer C
LEFT JOIN Transaction T ON C.CustomerID = T.CustomerID
GROUP BY C.CustomerID, CustomerName;
```

Query Result:

	CustomerID	CustomerName	LatestTransactionDate
▶	10	Tanasub Rattanaharirak	2023-11-18
	11	Poottapol Paannna	2023-11-28
	12	Phakphup Phoonphoem	2023-11-24
	13	Kasama Kiatapakul	2023-11-25
	14	Supasri Seekane	2023-11-26
	15	Nantipat Lnwza	NULL
	16	Pimmy Nekognum	2023-11-22
	17	Pathu Sudlhor	2023-11-27
	18	Napat Zaza	2023-11-23
	19	Ratchaneekorn Suaymak	NULL
	20	Thawanthong Lhormak	2023-11-21

28.Report: Product Category,Average Transaction Amount

Objective: Calculate the average transaction amount per product category.

SQL Command:

```
SELECT P.P_Category, AVG(T.Amount) AS AvgTransactionAmount
```

```
FROM Product P
```

```
JOIN Transaction T ON P.ProductID = T.ProductID
```

```
GROUP BY P.P_Category;
```

Query Result:

	P_Category	AvgTransactionAmount
►	Snack	300.000000
	Thai Desserts	150.000000
	Beverages	200.000000
	Fashion	300.000000
	Office Supplies	300.000000
	Sports	400.000000
	Books	450.000000
	Home Essentials	550.000000
	Toys	550.000000

29.Report: ProductID,Product Name

Objective: Find the product that have not been purchase yet.\

SQL Command:

```
SELECT P.ProductID, P.P_NAME
```

```
FROM Product P
```

```
LEFT JOIN Transaction T ON P.ProductID = T.ProductID
```

```
WHERE T.TransactionID IS NULL;
```

Query Result:

	ProductID	P_NAME
►	1132	Iphone 15 pro max 256GB
	1133	Ipad Pro 2022 256GB
	1134	Macbook Air M1 512

30.Report: CustomerID, Customer Name

Objective: List all customer who made transaction in the last 7 days

SQL Command:

```
SELECT DISTINCT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS  
CustomerName  
FROM Customer C  
JOIN Transaction T ON C.CustomerID = T.CustomerID  
WHERE T.Date >= CURDATE() - INTERVAL 7 DAY;
```

Query Result:

	CustomerID	CustomerName
▶	10	Tanasub Rattanaharirak
	13	Kasama Kiatapakul
	18	Napat Zaza
	20	Thawanthong Lhormak
	16	Pimmy Nekognum
	12	Phakphup Phoonphoem
	14	Supasri Seekane
	17	Pathu Sudlhor
	11	Poottapol Paannna

6588192 Supasri Seekane

Basic Queries

31.Report: ProductID, Product NAME, Product Description, Price, StockQuantity, Product Category

Objective: Retrieve all products with a price greater than 50

SQL Command:

```
SELECT *  
FROM Product  
WHERE Price > 50;
```

Query Result:

	ProductID	P_NAME	P_Description	Price	StockQuantity	P_Category
▶	1134	Macbook Air M1 512	Apple product	1599.99	10	Snack
	1133	Ipad Pro 2022 256GB	Apple product	1299.99	10	Snack
	1132	Iphone 15 pro max 256GB	Apple product	999.99	10	Snack
	1127	Shockproof Hat	Vibrant shockproof hat for sports	60.00	40	Sports
	1126	Genuine Leather Bag	Genuine leather bag made from selected animal ...	120.00	30	Fashion
	1125	Natural Ink Cartridge	Ink cartridge made from Thai herbs	80.00	50	Office Supplies
	1124	Traditional Thai Clothing	Handwoven traditional Thai clothing made from ...	150.00	20	Fashion
•	NULL	NULL	NULL	NULL	NULL	NULL

[illegible]

Advance Queries

34.Report: ProductID, Product NAME, TotalRevenue

Objective: Find the top 3 products with the highest total revenue

SQL Command:

```
SELECT P.ProductID, P.P_NAME, SUM(T.Amount) AS TotalRevenue
FROM Product P
JOIN Transaction T ON P.ProductID = T.ProductID
GROUP BY P.ProductID, P.P_NAME
ORDER BY TotalRevenue DESC
LIMIT 3;
```

Query Result:

	ProductID	P_NAME	TotalRevenue
	1129	Thai Tea Cup	500.00
▶	1130	Sensory Balls	550.00
	1131	Herbal Soap	600.00

35.Report: CustomerID, CustomerName

Objective: Retrieve customers who made transactions on weekdays (Monday to Friday)

SQL Command:

```
SELECT DISTINCT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS
CustomerName
FROM Customer C
JOIN Transaction T ON C.CustomerID = T.CustomerID
WHERE DAYOFWEEK(T.Date) BETWEEN 2 AND 6;
```

Query Result:

	CustomerID	CustomerName
▶	11	Poottapol Paannna
	12	Phakphup Phoonphoem
	16	Pimmy Nekognum
	17	Pathu Sudlhor
	18	Napat Zaza
	20	Thawanthong Lhormak

36.Report: ProductID, Product NAME, Product Description, Price, StockQuantity, Product Category

Objective: Retrieve products with a stock quantity less than the average stock quantity

SQL Command:

```
SELECT *  
FROM Product  
WHERE StockQuantity < (  
    SELECT AVG(StockQuantity)  
    FROM Product  
);
```

Query Result:

	ProductID	P_NAME	P_Description	Price	StockQuantity	P_Category
►	1124	Traditional Thai Clothing	Handwoven traditional Thai clothing made from ...	150.00	20	Fashion
	1125	Natural Ink Cartridge	Ink cartridge made from Thai herbs	80.00	50	Office Supplies
	1126	Genuine Leather Bag	Genuine leather bag made from selected animal ...	120.00	30	Fashion
	1127	Shockproof Hat	Vibrant shockproof hat for sports	60.00	40	Sports
	1132	Iphone 15 pro max 256GB	Apple product	999.99	10	Snack
	1133	Ipad Pro 2022 256GB	Apple product	1299.99	10	Snack
	1134	Macbook Air M1 512	Apple product	1599.99	10	Snack

37.Report: CustomerID, CustomerName, TotalSpending

Objective: Find the customer with the highest total spending

SQL Command:

```
SELECT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS CustomerName,  
SUM(T.Amount) AS TotalSpending  
FROM Customer C  
JOIN Transaction T ON C.CustomerID = T.CustomerID  
GROUP BY C.CustomerID, CustomerName  
ORDER BY TotalSpending DESC  
LIMIT 1;
```

Query Result:

	CustomerID	CustomerName	TotalSpending
►	11	Poottapol Paannna	600.00

38.Report: CustomerID, CustomerName

Objective: Find customers who made transactions on weekends (Saturday or Sunday)

SQL Command:

```
SELECT DISTINCT C.CustomerID, CONCAT(C.C_FNAME, ' ', C.C_LNAME) AS  
CustomerName  
FROM Customer C  
JOIN Transaction T ON C.CustomerID = T.CustomerID  
WHERE DAYOFWEEK(T.Date) IN (1, 7);
```

Query Result:

	CustomerID	CustomerName
▶	10	Tanasub Rattanaharirak
	13	Kasama Kiatapakul
	14	Supasri Seekane

39.Report: ProductID, Product NAME, Product Category, ProductRank

Objective: Retrieve the top-selling product for each product category

SQL Command:

```
WITH RankedProducts AS (  
    SELECT  
        P.ProductID,  
        P.P_NAME,  
        P.P_Category,  
        RANK() OVER (PARTITION BY P.P_Category ORDER BY COUNT(T.TransactionID)  
DESC) AS ProductRank  
    FROM Product P  
    LEFT JOIN Transaction T ON P.ProductID = T.ProductID  
    GROUP BY P.ProductID, P.P_NAME, P.P_Category  
)  
SELECT *  
FROM RankedProducts  
WHERE ProductRank = 1;
```

Query Result:

	ProductID	P_NAME	P_Category	ProductRank
▶	1123	Bubble Milk Tea	Beverages	1
	1128	Thai Language Book	Books	1
	1124	Traditional Thai Clothing	Fashion	1
	1126	Genuine Leather Bag	Fashion	1
	1129	Thai Tea Cup	Home Essentials	1
	1131	Herbal Soap	Home Essentials	1
	1125	Natural Ink Cartridge	Office Supplies	1
	1121	Supasri snack	Snack	1
	1127	Shockproof Hat	Sports	1
	1122	Taro Cupcake	Thai Desserts	1
	1130	Sensory Balls	Toys	1

40.Report: ProductID, Product NAME, MaxTransactionAmount, MinTransactionAmount

Objective: Find the products with the highest and lowest transaction amounts

SQL Command:

```
SELECT P.ProductID, P.P_NAME, MAX(T.Amount) AS MaxTransactionAmount,
```

```
MIN(T.Amount) AS MinTransactionAmount
```

```
FROM Product P
```

```
JOIN Transaction T ON P.ProductID = T.ProductID
```

```
GROUP BY P.ProductID, P.P_NAME;
```

Query Result:

	ProductID	P_NAME	MaxTransactionAmount	MinTransactionAmount
►	1121	Supasri snack	300.00	300.00
	1122	Taro Cupcake	150.00	150.00
	1123	Bubble Milk Tea	200.00	200.00
	1124	Traditional Thai Clothing	250.00	250.00
	1125	Natural Ink Cartridge	300.00	300.00
	1126	Genuine Leather Bag	350.00	350.00
	1127	Shockproof Hat	400.00	400.00
	1128	Thai Language Book	450.00	450.00
	1129	Thai Tea Cup	500.00	500.00
	1130	Sensory Balls	550.00	550.00
	1131	Herbal Soap	600.00	600.00